



GOSFORD HIGH SCHOOL

2010 EXTENSION 1 MATHEMATICS

Preliminary Course Assessment Task 2

PART 1

Special Instructions

- Students are to hand in their papers in three bundles.
- Questions 1 to 3 are to be in one bundle, Questions 4 to 6 in another and Questions 7 to 9 in another.
- Students must start Question 4 on a new page and Question 7 on a new page.

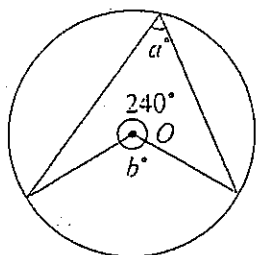
General Instructions

Time Allowed: 1 hour plus 5 minutes reading time

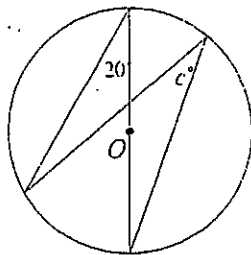
- Attempt all questions.
- Approved calculators may be used.
- Write using blue or black pen.
- Full marks may not be awarded where necessary working is not shown.

Question 1. Find the value of the pronumerals in each of the following.
(No reasons needed.)

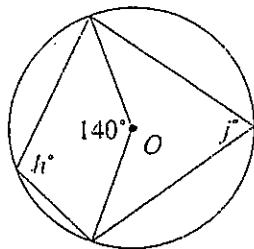
a



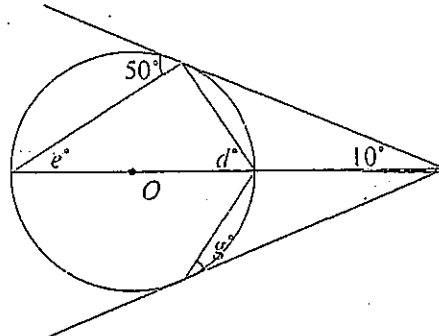
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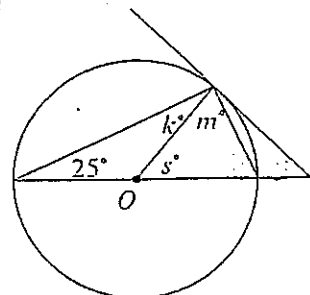
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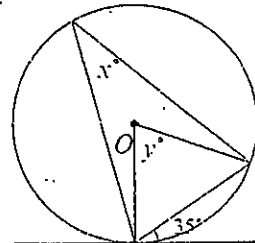
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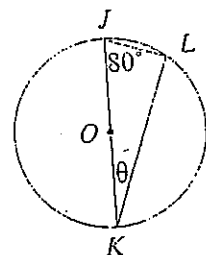
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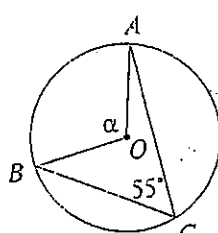
(13)

Question 2. Find the value of the pronumerals in each of the following giving reasons for your answers.

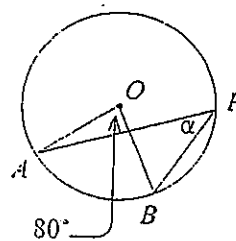
a



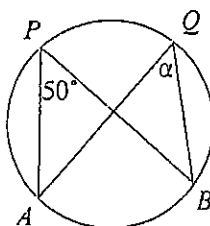
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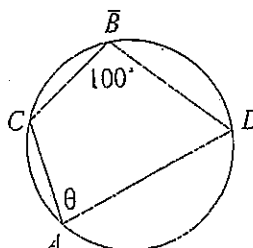
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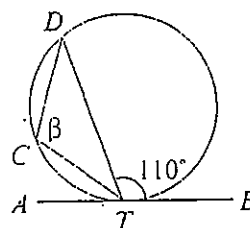
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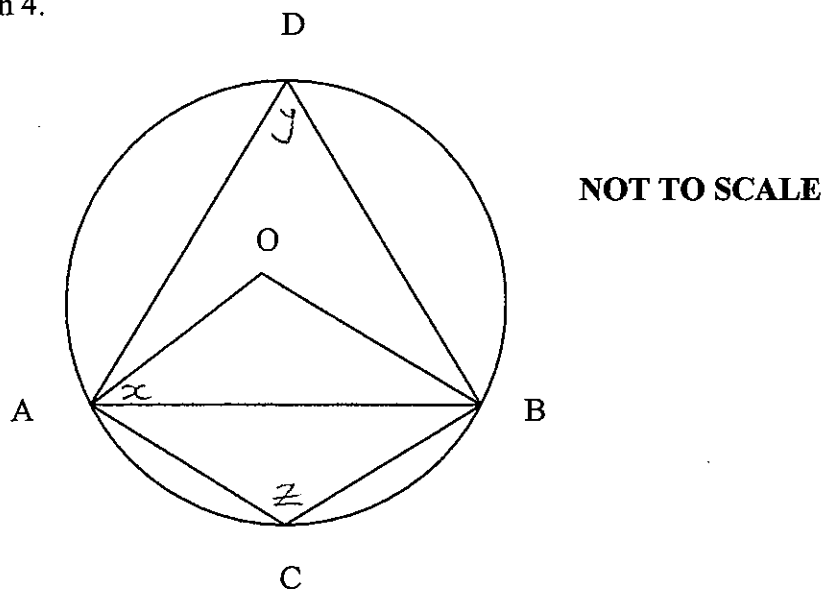
f



(12)

Question 3. Two parallel chords of length 50cm and 104cm are drawn in a circle; radius 65cm. How far apart are the chords? (5)

Question 4.



ABCD is a cyclic quadrilateral.

O is the centre of the circle. $\angle OAB = x^\circ$, $\angle ACB = z^\circ$ and $\angle ADB = y^\circ$.

If $\angle ACB$ is obtuse, prove that $z - y = 2x$.

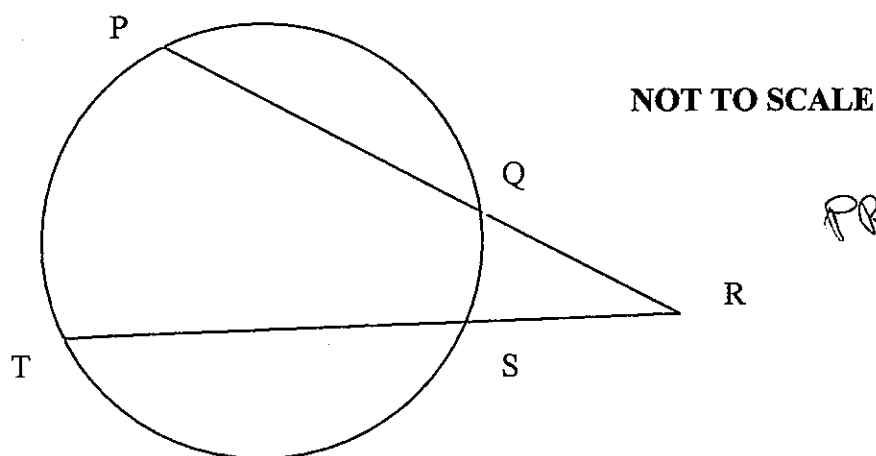
(4)

Question 5. AB is a chord of a circle and XAY is the tangent at A. AK and AL are chords bisecting $\angle XAB$ and $\angle YAB$ respectively.

(a) Draw a diagram clearly showing all of this information. (2)

(b) Prove that: (i) $AL = BL$. (ii) KL is a diameter of the circle. (4)

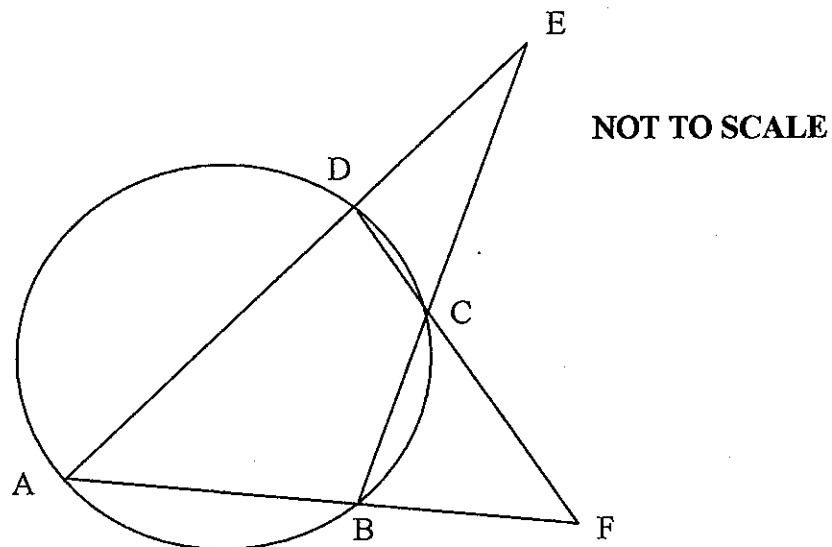
Question 6. PQ and TS are two secants of a circle intersecting externally at R. Given that $PQ = 7\text{cm}$, $QR = 5\text{cm}$ and $SR = 4\text{cm}$, find the length of TS.



$$PR \times QR = ($$

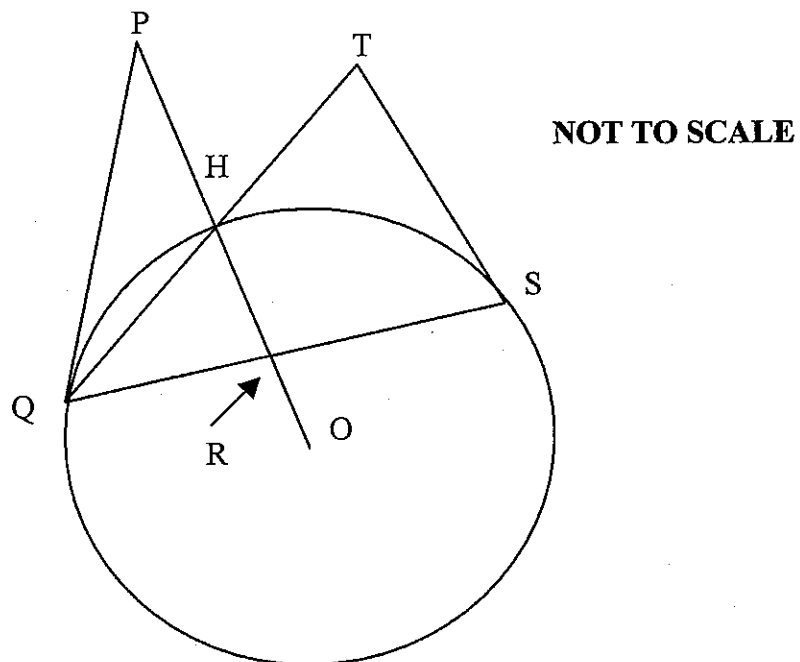
(2)

Question 7.



$ABCD$ is a cyclic quadrilateral. The angles AEB and AFD are equal.
Prove that AC is a diameter of the circle. (4)

Question 8.



PQR is an isosceles triangle such that $PQ = PR$. PQ is a tangent at Q to the circle centre O . PR produced passes through O . QR produced meets the circle at S . The tangent to the circle at S meets QH produced at T .

Prove that angle HQR is 45° . (4)

Question 1:

a) $a = \cancel{120} 60^\circ$
 $b = 120^\circ$

b) $c = 20^\circ$

c) $j = \cancel{140} 70^\circ$
 $h = 110^\circ$

d) $d = 50^\circ$
 $e = 40^\circ$

e) $q = 40^\circ$
 $r = 25^\circ$
 $m = 65^\circ$
 $s = 50^\circ$

f) $x = 35^\circ$
 $y = 70^\circ$

Question 2

a) $\angle JLK = 90^\circ$ (angle in a semicircle is 90°)
 $\theta + 80 + 90 = 180$ (angle sum of $\triangle JKL$ is 180°)
 $\therefore \theta = 10$

b) $\angle x = 110^\circ$ (angle at the centre is twice the angle at the circumference standing on the same arc.)

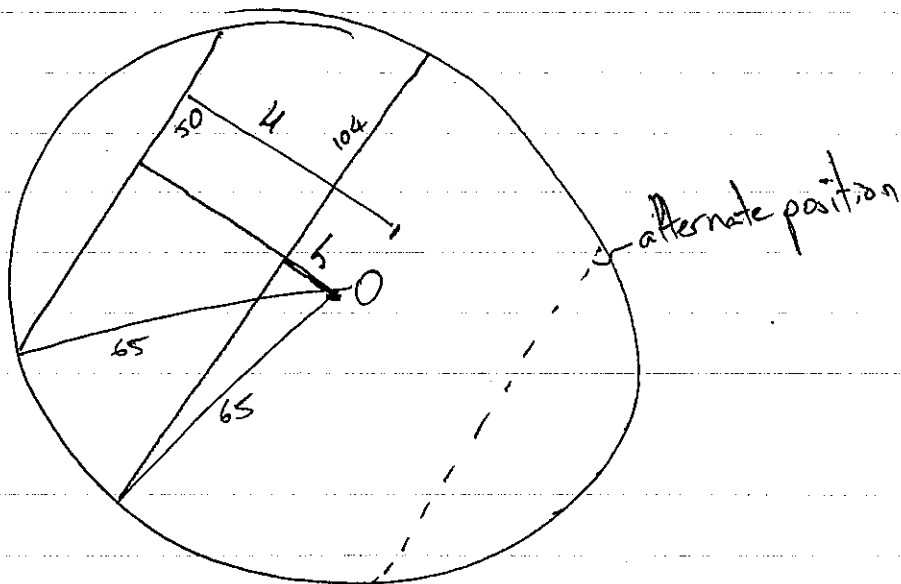
c) $\angle x = 40^\circ$ (angle at the centre is twice the angle at the circumference standing on the same arc.)

d) $\angle x = 50^\circ$ (angles in the same segment are equal.)

e) $\theta = 80^\circ$ (opposite $\angle$'s in a cyclic quadrilateral are supplementary.)

f) $\beta = 110^\circ$ (angle between a chord & a tangent is equal to the angle in the alternate segment.)

Question 3



$$H^2 + 25^2 = 65^2$$

$$H^2 =$$

$$H =$$

$$52^2 + h^2 = 65^2$$

$$h^2 =$$

$$h =$$

$\therefore$ distance is either

$$H - h \quad \text{or} \quad H + h$$

$$= \quad =$$

Question 4

~~Given~~ $AO = BO$ (equal radii)

$\therefore \triangle ABO$ is isosceles

$\therefore \angle ABO = x$ (base $\angle$'s in isosceles $\triangle$'s are equal)

$\angle AOB = 180 - 2x$ (angle sum of $\triangle AOB$ is 180°)

$\angle AOB = 2 \times \angle ADB$ (angle at the centre is twice the angle at the circumference standing on the same arc)

$$180 - 2x = 2y$$

Also $\angle ACB = 180 - \angle ADB$ (opposite $\angle$'s in a cyclic quad are supplementary)

$$\therefore z = 180 - y$$

$$y = 180 - z$$

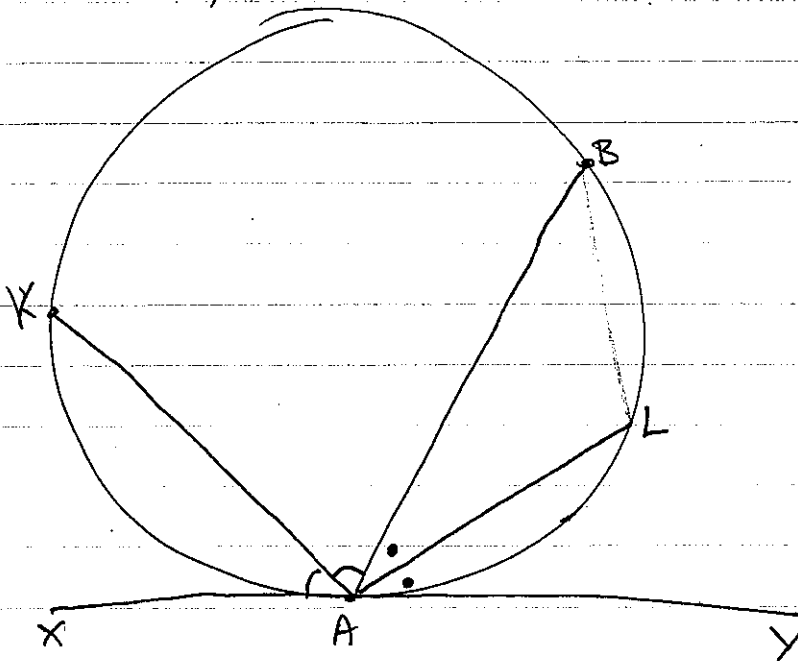
by substitution

$$180 - 2x = y + 180 - z$$

$$z - y = 2x$$

Question 5

a)



b) i) $\angle ABL = \angle BAL =$
~~Given~~ $\angle BAL = \angle LAY$ (given)
 $\angle ABL = \angle LAY$ (angle between a tangent & a chord is equal to the angle in the alternate segment)

$\therefore \triangle ABL$ is isosceles

$$(\angle BAL = \angle ABL)$$

$$\therefore AL = BL$$

ii) $\angle KAX = \angle BAK$ & $\angle BAL = \angle LAY$ (given)

~~$2\angle KAB + 2\angle LB$~~

$2\angle KAB + 2\angle BAL = 180$ (adjacent supplementary)

$\therefore \angle KAB + \angle BAL = 90$

$\angle KAL = 90$

Since angle at the circumference is 90°
 KL must be a diameter.

Question 6

$PR \times QR = TR \times SR$

(ratio of the intercepts of a chord is maintained)

Let $TS = x$

$(7+5) \times 5 = (x+4) \times 4$

$60 = 4x + 16$

$4x = 44$

$x = 11$

Question 7

$$\angle ADF = 180 - \angle DAF - \angle AFD \quad (\text{angle sum of } \triangle AFD \text{ is } 180^\circ)$$

similarly $\angle AEB = 180 - \angle DAF - \angle AEB$

since $\angle AEB = \angle AFD$ (given)

$$\angle AEB = 180 - \angle DAF - \angle AFD$$

$$\therefore \angle ADC = \angle ABC$$

$$\angle ADC + \angle ABC = 180^\circ \quad (\text{opposite } \angle\text{'s in a cyclic quad are supplementary})$$

$$\therefore 2\angle ADC = 180^\circ$$

$$\angle ADC = 90^\circ$$

$$\therefore AC \text{ is a diameter} \quad (\text{angle in a semi-circle is } 90^\circ)$$

Question 8)

$$\angle PQH = \angle HSQ$$

$$\angle HSQ = \frac{1}{2} \angle QOH$$

$$\angle PQO = 90^\circ$$

(angle between chord..

(angle at the centre

(angle between a radius & a tangent

$$\text{Let } \angle HQR = x$$

$$\angle PQR = 90 - 2\angle HSR$$

$$\angle QPR + 2\angle HQR = 180$$

$$90 - 2\angle HSR + 2\angle HQR = 180$$

$$\angle HQR = \frac{45}{2} + \angle HSR$$

$$\angle HSR + \angle HQR = 45 + \angle HSR$$



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PART 2

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- Students must start **PART B** on a new page.

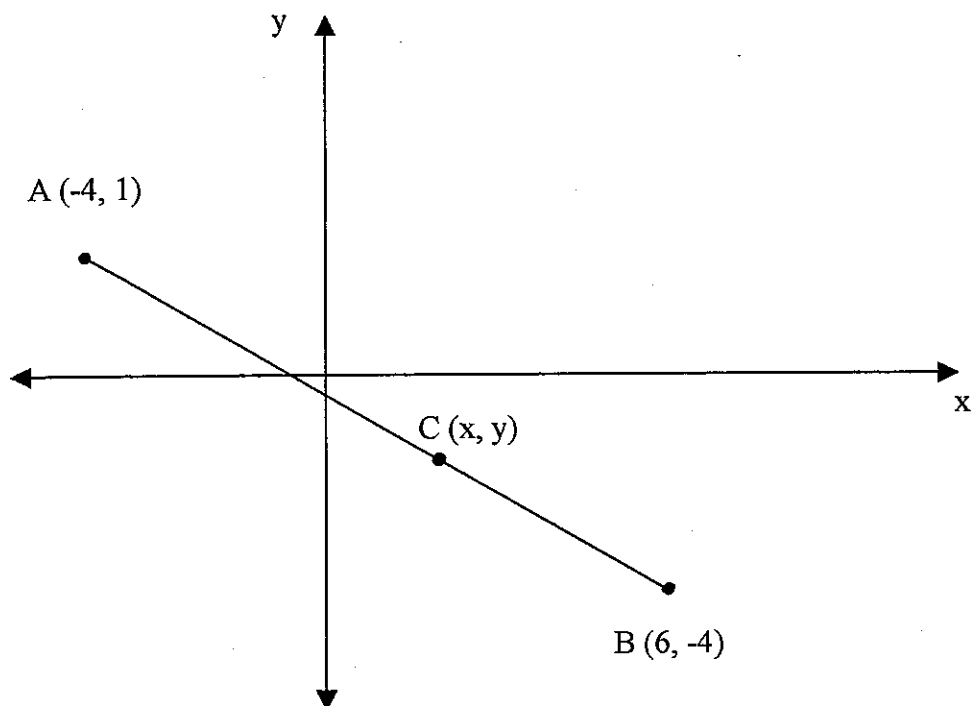
General Instructions

Time Allowed: 1 hour plus 5 minutes reading time

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PART A.

1.



Given that $2AC = 3BC$, find the coordinates of the point C. (3)

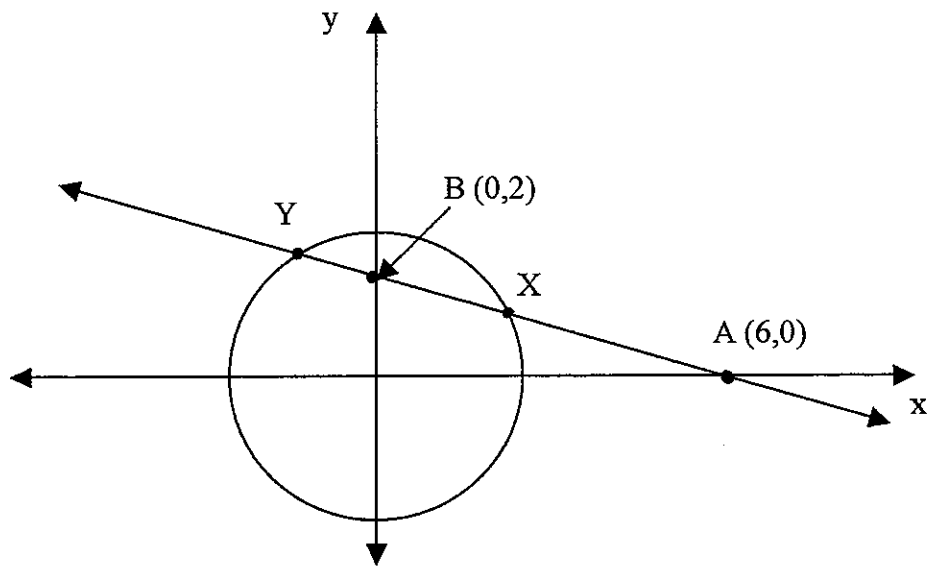
2. Find the coordinates of the point P, which divides the interval joining A (-3,6) and B (1,10) externally in the ratio 5 : 3. (3)

3. The point C (-1, -4) divides the interval AB internally in the ratio 2 : 1. If the coordinates of A are (3,2), find the coordinates of B. (3)

4. A (6,0) and B (0,2) are points on a number plane.

(a) Find the coordinates of the point C which divides the interval AB in the ratio $k : 1$. (1)

(b) Hence, or otherwise, find the ratio in which Y divides the interval AB, given that the line AB meets the circle $x^2 + y^2 = 10$ at X and Y. (6)



PART B. START A NEW PAGE

1. Sketch the graph of $y = x(x+1)(x-1)^2$ (2)

2. Consider the polynomial $P(x) = 2x^3 + kx^2 - 18x - 8$.

(i) Find the value of k for which $(x+2)$ is a factor of $P(x)$. (1)

(ii) For this value of k express $P(x)$ as a product of linear factors. (2)

3. Solve $x^3 + 2x^2 - 5x - 6 = 0$. (3)

4. If α , β and γ are the roots of the equation $x^3 - 5x^2 - 3x + 2 = 0$, find the value of

(i) $\alpha + \beta + \gamma$. (1)

(ii) $\alpha\beta\gamma$. (1)

(iii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. (2)

(iv) $\alpha^2\beta^2 + \beta^2\gamma^2 + \alpha^2\gamma^2$. (2)

5. The polynomial $P(x) = x^3 - 8x^2 + kx - 12$ has one root equal to the sum of the other two roots. Find the value of k . (3)

6. Two of the roots of $x^3 - 15x + 4 = 0$ are reciprocals of one another. Find all three roots. (3)

7. When $ax^3 + bx^2 + cx + d$ is divided by $x - \alpha$ the remainder is r_1 , and when it is divided by $x + \alpha$ the remainder is r_2 .

Prove that $\alpha = \frac{b(r_1 - r_2)}{a(r_1 + r_2) + 2(bc - ad)}$. (4)

PART 2

SOLUTIONS

PART A

$$1. \quad \text{If } 2AC = 3BC$$

$$\frac{AC}{BC} = \frac{3}{2}$$

$$(-4, 1) \quad (6, -4)$$

$$3:2$$

$$x = \frac{2(-4) + 3(6)}{5}$$

$$y = \frac{2(1) + 3(-4)}{5}$$

$$x = 2$$

$$y = -2$$

$$C \text{ is } (2, -2)$$

(3)

$$2. \quad (-3, 6) \quad (1, 10)$$

$$5:-3$$

$$x = \frac{-3(-3) + 5(1)}{2}$$

$$y = \frac{-3(6) + 5(10)}{2}$$

$$x = -2$$

$$y = 16$$

$$P \text{ is } (-2, 16)$$

(3)

$$3. \quad (3, 2) \quad (x_1, y_1)$$

$$2:1$$

$$-1 = \frac{1(3) + 2(x_1)}{3}$$

$$-4 = \frac{1(2) + 2(y_1)}{3}$$

$$-3 = 3 + 2x_1$$

$$-12 = 2 + 2y_1$$

$$2x_1 = -6$$

$$2y_1 = -14$$

$$x_1 = -3$$

$$y_1 = -7$$

$$B \text{ is } (-3, -7)$$

(3)

$$4. \quad a) \quad (6, 0) \quad (0, 2)$$

$$k:1$$

$$x = \frac{1(6) + k(0)}{k+1}$$

$$y = \frac{1(0) + k(2)}{k+1}$$

$$1. \quad \frac{6}{k+1}$$

$$= \frac{2k}{k+1}$$

$$\left(\frac{6}{k+1}, \frac{2k}{k+1} \right)$$

(1)

b) Eqn of AB is $y = 2 - \frac{x}{3}$

(1)

$$x^2 + y^2 = 10 \quad \text{--- (1)}$$

$$y = 2 - \frac{x}{3} \quad \text{--- (2)}$$

$$\therefore x^2 + \left(2 - \frac{x}{3}\right)^2 = 10$$

$$\therefore x^2 + 4 - \frac{4x}{3} + \frac{x^2}{9} = 10$$

$$9x^2 + 36 - 12x + x^2 = 90$$

$$10x^2 - 12x - 54 = 0$$

$$5x^2 - 6x - 27 = 0$$

$$5x \quad \times \quad 9$$

$$x \quad \times \quad -3$$

$$(5x + 9)(x - 3) = 0$$

$$x = -\frac{9}{5} \quad \text{or} \quad 3$$

(2)

$$\therefore \text{if } x = -\frac{9}{5}, y = 2 - \frac{-\frac{9}{5}}{3}$$

$$= \frac{13}{5}$$

(1)

$$Y \Rightarrow \left(-\frac{9}{5}, \frac{13}{5}\right)$$

From a) $\frac{b}{k+1} = -\frac{a}{5}$

$$30 = -9k - 9$$

$$-9k = -39$$

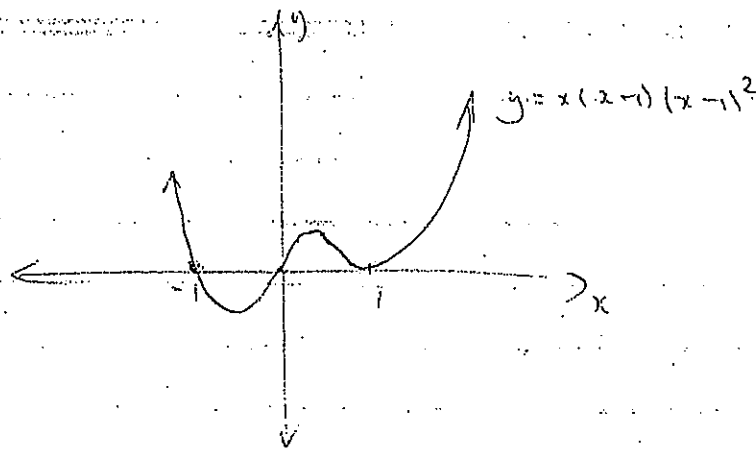
$$k = \frac{13}{3}$$

(2)

$\therefore Y$ divides AB externally in the ratio 13 : 3. (1)

PART B

1.



(2)

2. (i) $P(x) = 2x^3 + kx^2 - 18x - 8$

$P(-2) = 2(-2)^3 + k(-2)^2 - 18(-2) - 8$

$0 = -16 + 4k + 36 - 8$

$4k = -12$

$k = -3$

(1)

(ii)

$$\begin{array}{r}
 2x^2 - 7x - 4 \\
 2+2 \overline{) 2x^3 - 3x^2 - 18x - 8} \\
 \underline{2x^3 + 4x^2} \\
 -7x^2 - 18x \\
 \underline{-7x^2 - 14x} \\
 -4x - 8 \\
 \underline{-4x - 8} \\
 0
 \end{array}$$

$\therefore P(x) = (x+2)(2x^2 - 7x - 4)$
 $= (x+2)(2x+1)(x-4)$

(2)

3. Let $P(x) = x^3 + 2x^2 - 5x - 6$

$P(1) = 1 + 2 - 5 - 6$
 $\neq 0$

$P(-1) = -1 + 2 + 5 - 6$
 $= 0$

$(x+1)$ is a factor

(1)

$$\begin{array}{r}
 x^2 + x - 6 \\
 x+1 \overline{) x^3 + 2x^2 - 5x - 6} \\
 \underline{x^3 + x^2} \\
 x^2 - 5x - 6 \\
 \underline{x^2 + x} \\
 -6x - 6 \\
 \underline{-6x - 6} \\
 0
 \end{array}$$

$$\therefore f(x) = (x+1)(x+3)(x-2)$$

$$\therefore x = -1, -3 \text{ or } 2$$

(2)

$$4. \quad (i) \quad \alpha + \beta + \gamma = -\frac{b}{a} = 5$$

(1)

$$(ii) \quad \alpha\beta\gamma = -\frac{c}{a} = -2$$

(1)

$$\begin{aligned}
 (iii) \quad \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma} \\
 &= \frac{c/a}{-c/a} \\
 &= 3/2
 \end{aligned}$$

(2)

$$\begin{aligned}
 (iv) \quad (\alpha\beta + \beta\gamma + \alpha\gamma)^2 &= [\alpha\beta + \beta\gamma]^2 + 2[\alpha\beta + \beta\gamma]\alpha\gamma + (\alpha\gamma)^2 \\
 &= \alpha^2\beta^2 + 2\alpha\beta^2\gamma + \beta^2\gamma^2 + 2\alpha^2\beta\gamma + 2\alpha\beta\gamma^2 + \alpha^2\gamma^2 \\
 &= \alpha^2\beta^2 + \beta^2\gamma^2 + \alpha^2\gamma^2 + 2\alpha\beta\gamma(\alpha + \beta + \gamma)
 \end{aligned}$$

$$\begin{aligned}
 \therefore \alpha^2\beta^2 + \beta^2\gamma^2 + \alpha^2\gamma^2 &= (\alpha\beta + \beta\gamma + \alpha\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma) \\
 &= (-3)^2 - 2 \times -2(5) \\
 &= 9 + 20 \\
 &= 29
 \end{aligned}$$

(2)

5. Let the roots be α, β, δ

$$\therefore \alpha = \beta + \delta$$

$$\text{Hence } \alpha + \alpha = -\frac{b}{a}$$

$$2\alpha = 8$$

$$\alpha = 4$$

$$\therefore P(4) = 0$$

(3)

$$(4)^3 - 8(4)^2 + 4k - 12 = 0$$

$$64 - 128 + 4k - 12 = 0$$

$$4k = 76$$

$$k = 19$$

6. Let the roots be $\alpha, \beta, \frac{1}{\beta}$

$$\text{Now } \alpha, \beta, \frac{1}{\beta} = -\frac{a}{a}$$

$$\alpha = -4$$

(1)

$$\text{Also } \alpha + \beta + \frac{1}{\beta} = -\frac{b}{a}$$

$$-4 + \beta + \frac{1}{\beta} = 0$$

$$\beta + \frac{1}{\beta} = +4$$

$$\beta^2 + 1 = +4\beta$$

$$\beta^2 - 4\beta + 1 = 0$$

$$\beta = \frac{+4 \pm \sqrt{16-4}}{2}$$

$$= \frac{+4 \pm 2\sqrt{3}}{2}$$

$$= +2 \pm \sqrt{3}$$

(2)

$\therefore$ Roots are $4, (+2+\sqrt{3}), (+2-\sqrt{3})$

$$7. \quad a x^3 + b x^2 + c x + d = r_1 \quad \text{--- (1)}$$

$$- a x^3 + b x^2 - c x + d = r_2 \quad \text{--- (2)}$$

$$(1) + (2) \Rightarrow 2b x^2 + 2d = r_1 + r_2$$

$$x^2 = \frac{r_1 + r_2 - 2d}{2b}$$

$$(1) - (2) \Rightarrow 2a x^3 + 2c x = r_1 - r_2$$

$$2x(a x^2 + c) = r_1 - r_2$$

$$x = \frac{r_1 - r_2}{2(a x^2 + c)}$$

$$\text{But } x^2 = \frac{r_1 + r_2 - 2d}{2b}$$

$$\therefore x = \frac{r_1 - r_2}{2(a \left[\frac{r_1 + r_2 - 2d}{2b} \right] + c)}$$

$$= \frac{r_1 - r_2}{2(a \left[\frac{r_1 + r_2 - 2d}{2b} \right] + 2bc)}$$

$$= \frac{2b(r_1 - r_2)}{2(a(r_1 + r_2) - 2ad + 2bc)}$$

$$= \frac{b(r_1 - r_2)}{a(r_1 + r_2) - 2(bc - ad)}$$

$$= \frac{b(r_1 - r_2)}{a(r_1 + r_2) - 2(bc - ad)}$$

$$= \frac{b(r_1 - r_2)}{a(r_1 + r_2) - 2(bc - ad)}$$

4